
Emergency Department & Patient Flow Analytics

Why a 90% access target was hiding a clinical-risk problem

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SQL (star schema, CTEs, window functions) · Python (pandas, NumPy, matplotlib) · Power BI + DAX

github.com/liuyizhou0402/data-analysis / [healthcare-ed-analytics](#)

54,837

ED presentations

11,045

inpatient admissions

14

analysis figures

7

SQL business queries

An end-to-end analytics project built around the metrics Australian health services actually report — the NEAT 4-hour target, ATS triage performance, left-without-being-seen, and 28-day readmissions. It runs the full path a health data analyst owns: data model, SQL, exploratory analysis, dashboard, and costed recommendations a non-technical executive can act on. The analytical point of the project is that the department's headline KPI is **arithmetically true and operationally misleading**, and the work shows how to prove that and what to do about it.

Three headline insights

- **The sickest patients wait the longest.** Overall 4-hour (NEAT) performance is 90%, but that blends away the clinical risk: T1 *Resuscitation* patients meet the target only **52.6%** of the time versus **99.4%** for non-urgent T5.
- **Winter demand is predictable and under-resourced.** Jun–Aug presentations run ~50% higher and 4-hour performance drops **5.2 percentage points** — a forecastable surge, not a surprise.
- **A small cohort drives outsized demand.** The top **1% of patients (107 people)** account for **4.9% of all presentations**, averaging 25 visits and 6.3 chronic conditions each — a clear care-coordination target that also moves the readmission rate.

On the data

The dataset is **synthetic and seed-reproducible** (`generate_data.py`), modelled on publicly reported Australian ED patterns. No real patient records are used, and no conclusion here describes a real health service. Synthetic data is a deliberate choice: patient-level ED data cannot be published, so the project demonstrates method on data that can be.

The finding, in one page

The department meets the 4-hour National Emergency Access Target for **90% of presentations**. Reported as a single blended number, that reads as a department comfortably in control. Disaggregated by triage category, it inverts:

Triage	Acuity	% within 4 hr	Avg ED LOS (min)	Admission rate
T1	Resuscitation	52.6%	236	~79%
T2	Emergency	67.8%	212	—
T3	Urgent	85.1%	178	—
T4	Semi-urgent	97.2%	129	—
T5	Non-urgent	99.4%	96	~3%

The 90% is carried by low-acuity patients who are fast to process and rarely admitted. The clinically urgent T1–T2 cohort — where delay carries real risk — meets the target only **66.5%** of the time. The blended KPI does not merely lose detail; it moves in the opposite direction to the risk it is meant to govern.

So what

And the cause is not where the KPI points. Admission likelihood runs from ~79% at T1 to ~3% at T5, and median ED length of stay rises monotonically with acuity. The patients breaching four hours are the ones waiting for an *inpatient bed*. This is access block, not triage speed — so the intervention belongs on the ward, not the front door. Three further findings (a forecastable winter surge, an LWBS rate above target, and a 1% cohort driving 5% of demand) are developed in the report that follows.

Contents

- **Summary dashboard** — the whole department on one page
- **1. Demand** — volume, seasonality, acuity mix, and the hour × day signature
- **2. Access & timeliness** — the 4-hour target, and where it breaks
- **3. Arrival patterns** — matching rosters to the curve
- **4. Quality** — patients who leave without being seen
- **5. Admissions & inpatient flow** — the bed-access bottleneck
- **6. Readmissions** — 28-day, by diagnosis group
- **7. High utilisers** — concentration of demand
- **8. Recommendations** — four costed actions and a KPI scorecard
- **Appendix** — data model, SQL techniques, reproduction steps, limitations

Summary dashboard

Rendered from the dataset by `analysis/build_dashboard.py` (Python + matplotlib). A step-by-step Power BI build guide with all DAX measures reproduces the same views as an interactive `.pbix` — the repository carries both so the work is portable to whichever BI stack a team already runs.

Emergency Department — Performance & Patient Flow

Synthetic dataset · FY2024-25 · 54,837 presentations · built with Python + matplotlib



Synthetic data for portfolio demonstration — no real patient records. Metrics modelled on publicly reported Australian ED indicators.

1. Demand — how much, and when

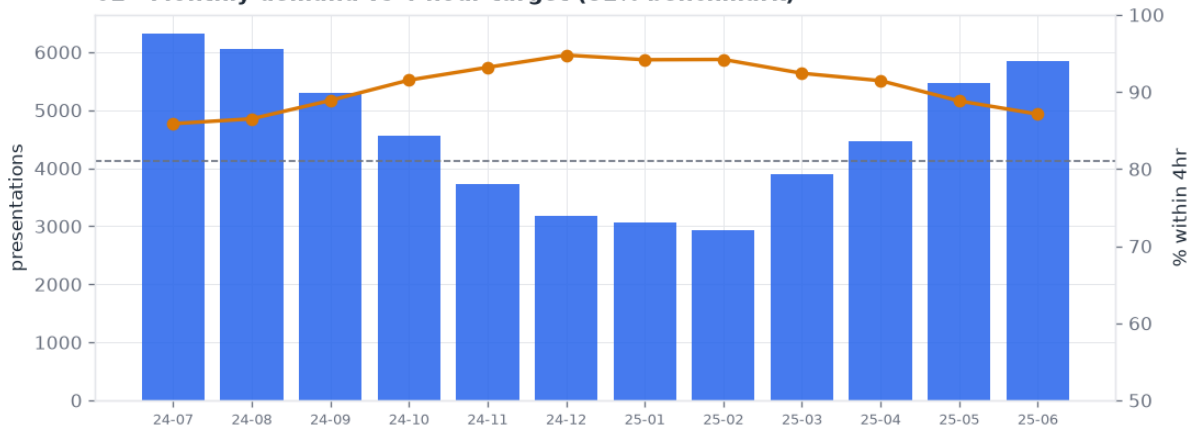
01 • Daily ED presentations — winter surge is clearly visible



Daily presentations with a 7-day moving average.

Presentations average ~150/day but swing seasonally. The 7-day average makes a **winter surge (Jun–Aug)** unmistakable — demand rises roughly 50% above the summer baseline. Because it is seasonal and recurring, it is *plannable*.

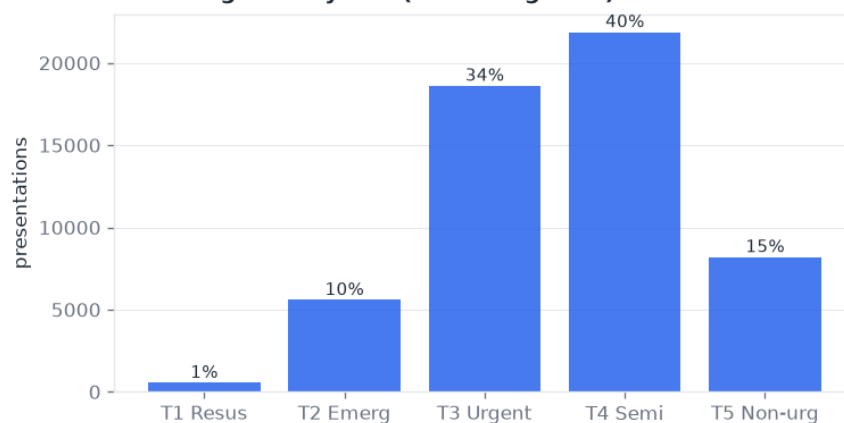
02 • Monthly demand vs 4-hour target (81% benchmark)



Monthly volume against 4-hour target performance.

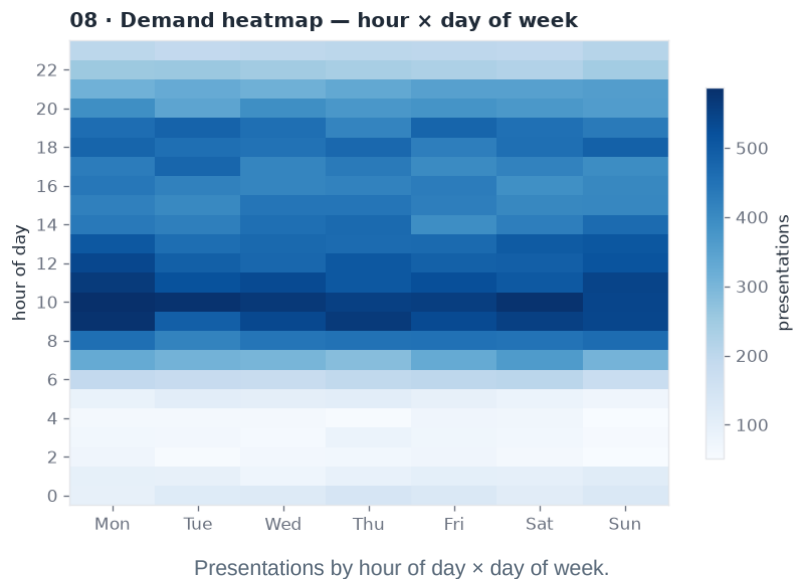
Overlaying monthly volume with 4-hour performance shows the two move in opposition: as winter volume climbs, the share of patients seen within four hours falls below the 81% benchmark. Demand is the lever on performance — which is what makes the surge worth pre-empting rather than absorbing.

03 • Triage acuity mix (ATS categories)



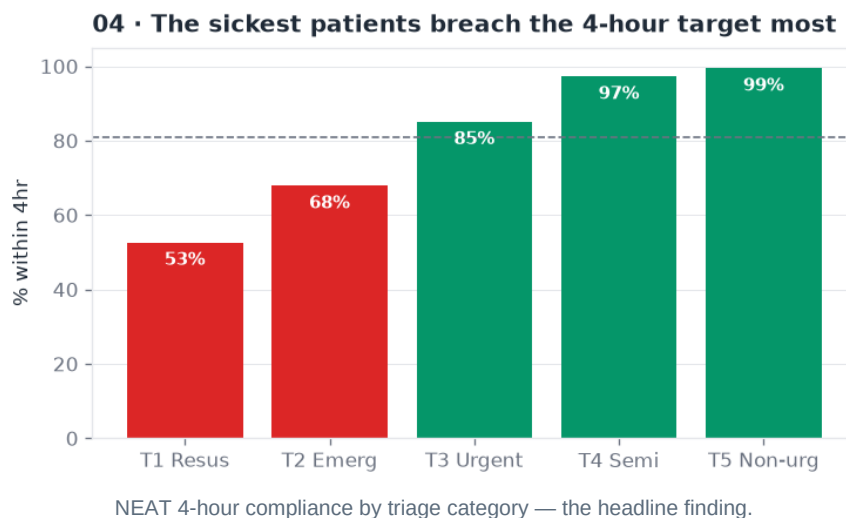
Presentation mix by ATS triage category.

The acuity mix is typical of a mixed metropolitan ED: **~75% of presentations are lower-acuity (T4–T5)**, while the smaller high-acuity group (T1–T2, ~11%) drives most of the clinical risk and nearly all inpatient demand. That asymmetry is exactly why a volume-weighted KPI flatters the department.

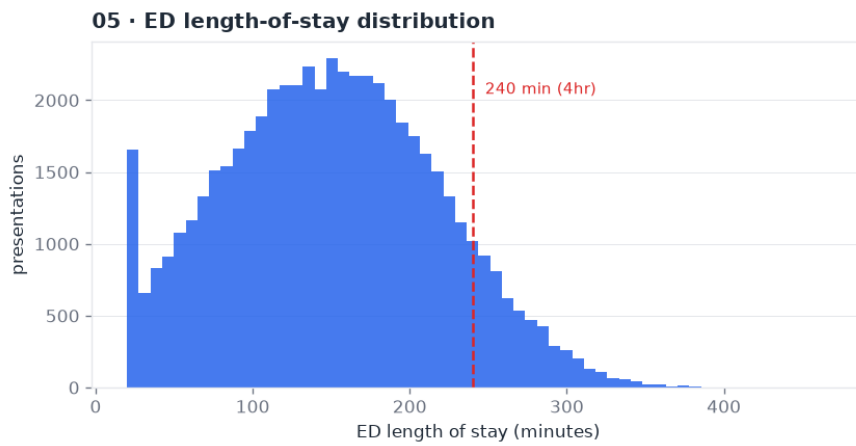


Demand concentrates **09:00–20:00 on weekdays**, peaking mid-morning; overnight (00:00–06:00) is consistently quiet. This hour × day signature is the basis for demand-matched rostering in §3.

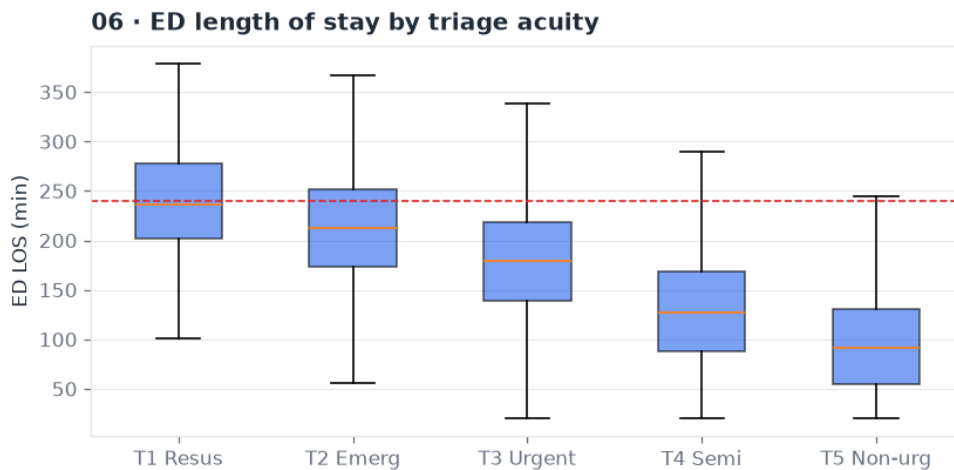
2. Access & timeliness — the 4-hour target



The headline finding, restated against the chart. Overall compliance is 90%; T1 resuscitation patients meet it only **52.6%** of the time, against 99.4% for non-urgent T5. Reported as one number, the KPI hides a clinical-risk problem in precisely the patients who matter most.



Most stays cluster under the 240-minute line, but a meaningful right tail breaches it. Those breaches are not randomly distributed — they concentrate in high-acuity, admitted patients.

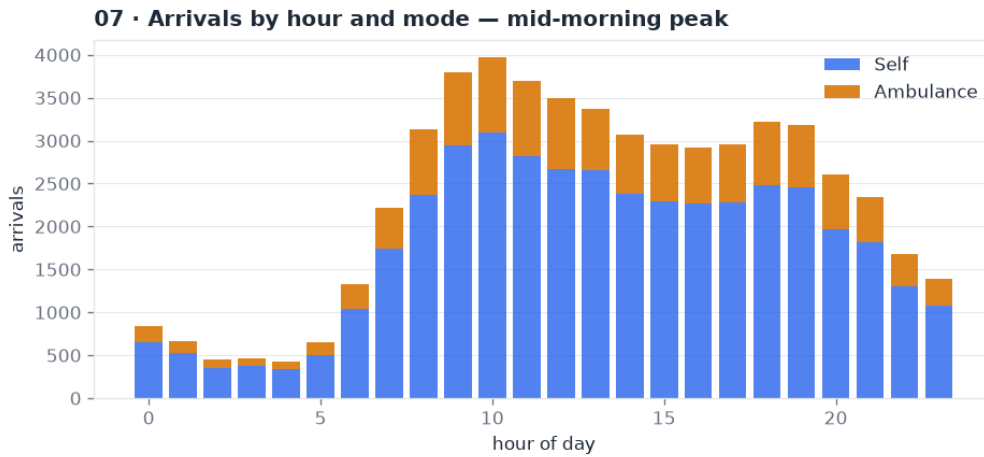


Median ED length of stay rises monotonically with acuity: T1–T2 patients sit near or beyond the 4-hour line while T4–T5 clear quickly. Since high-acuity patients are also the most likely to be admitted (§5), this points to **access block — inpatient bed availability** — rather than triage speed as the binding constraint.

Why this matters for the recommendation

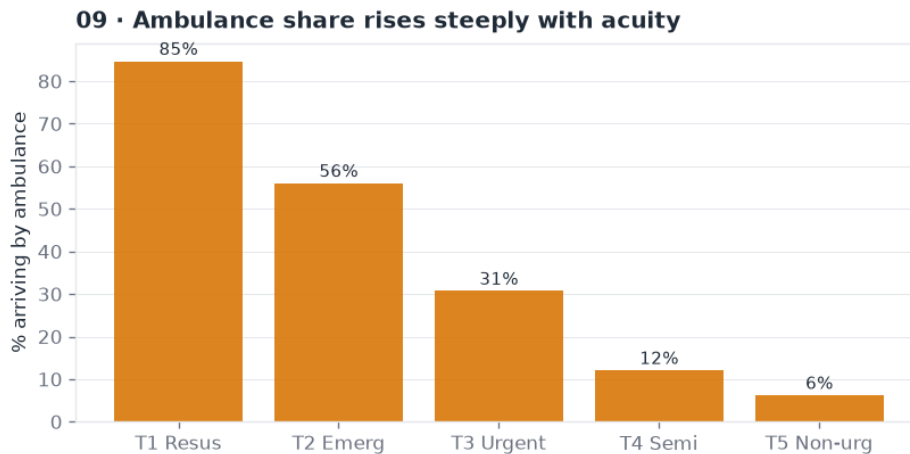
This is the analytical hinge of the project. Two facts that each look like an ED performance story — high-acuity patients breach, and high-acuity patients stay longest — only resolve once the admission rate is brought alongside them. The bottleneck sits downstream of the department being measured. An ED-side intervention (more triage nurses, faster streaming) would spend real money against the wrong constraint.

3. Arrival patterns & staffing



Arrivals by hour of day.

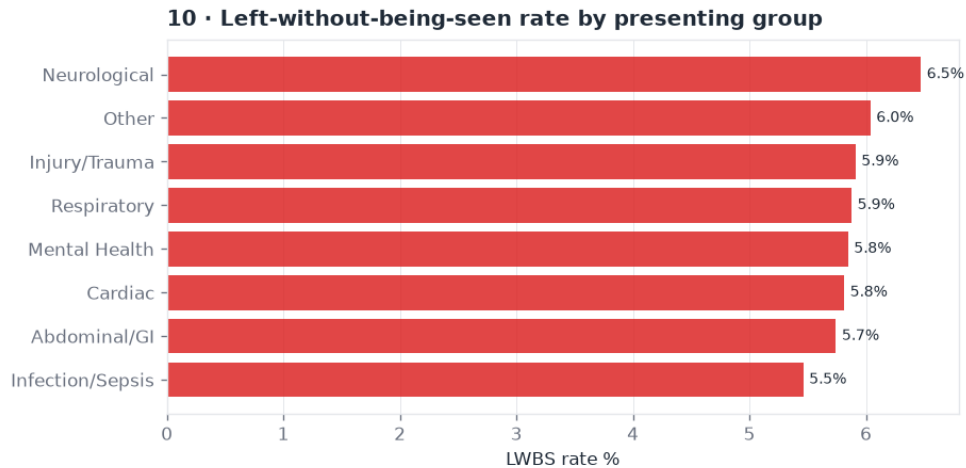
Arrivals ramp from ~07:00, peak **09:00–13:00** (top hour ~3,970 arrivals), and taper through the evening. Rostering senior decision-makers to this curve — rather than to a flat shift pattern — is the cheapest available lever on flow, because it requires no additional headcount.



Ambulance arrival share by triage category.

Ambulance share climbs steeply with acuity: **~85% of T1 arrivals** come by ambulance versus a small fraction of T5. Ambulance arrival is therefore a usable real-time proxy for incoming high-acuity load — and, given \$2, for ramping and access-block risk a few hours ahead.

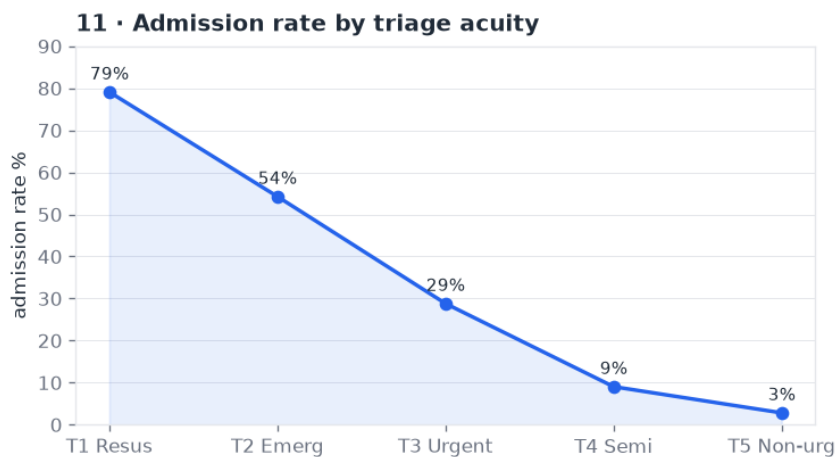
4. Quality — patients who leave



Left-without-being-seen rate by presenting group.

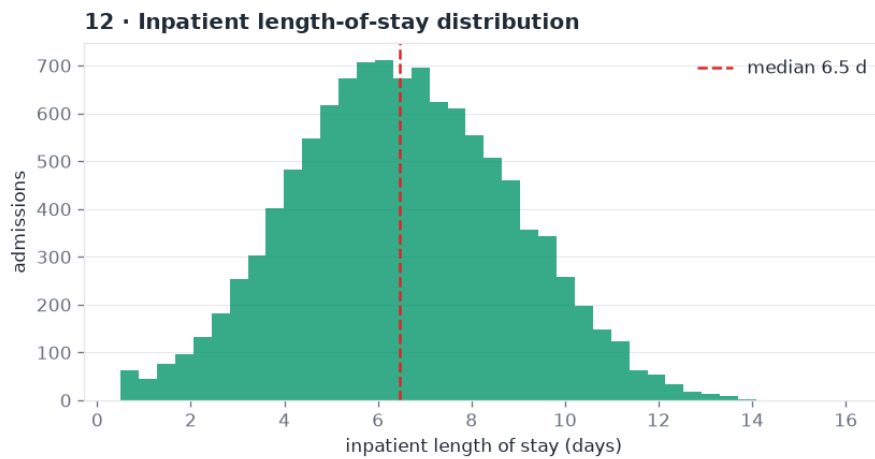
Left-without-being-seen (LWBS) averages **5.9%**, above the <5% quality target. It concentrates in lower-acuity presenting groups (Neurological, Other, Injury) that wait longest when the department is busy. LWBS is both a patient-safety signal and an access-equity one: the patients absorbing the queue are the patients the blended KPI counts as successes.

5. Admissions & inpatient flow



Probability of admission by triage category.

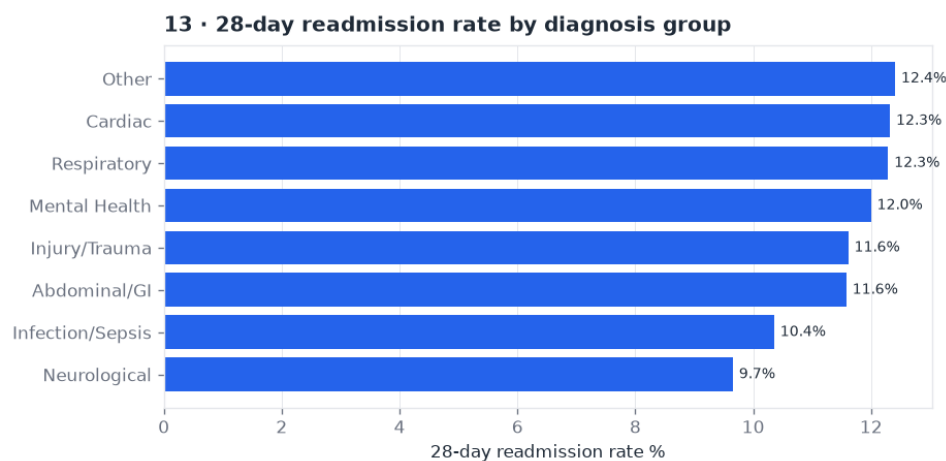
Admission likelihood scales sharply with acuity — **~79% for T1 down to ~3% for T5**. This is the mechanism behind §2: high-acuity patients breach four hours because they need an inpatient bed, and that depends on ward discharge flow, not on the ED front-door.



Inpatient length-of-stay distribution.

Inpatient length of stay is right-skewed with a **median of 6.5 days** and a long tail of complex, longer-stay patients — the cohort where discharge planning and step-down capacity pay off most, and the lever that actually moves the T1–T2 access figure.

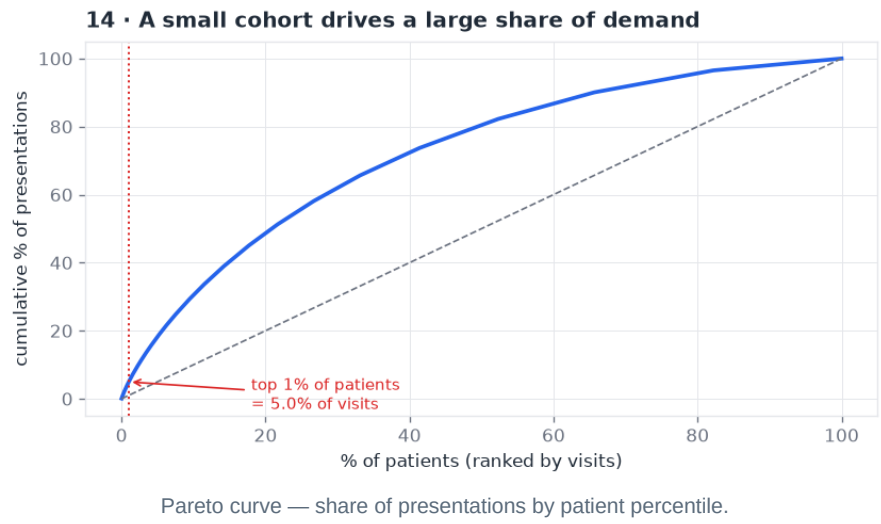
6. Readmissions



28-day readmission rate by diagnosis group.

28-day readmission averages **11.7%**, highest in **Cardiac and Respiratory (~12.3%)** — the classic chronic-disease groups where discharge quality, follow-up and community care determine whether a patient returns.

7. High utilisers



Demand is concentrated: the **top 1% of patients (107 people)** account for **4.9% of all presentations**, averaging 25 visits and 6.3 chronic conditions each. This is the same high-comorbidity cohort that drives the readmission rate in §6 — so a single care-coordination intervention moves two KPIs at once, which is what makes it the cheapest of the four recommendations to justify.

What the whole analysis says

Theme	Evidence	So what
Blended KPI hides risk	T1 4-hr = 52.6% vs 90% overall	Report NEAT by triage
Winter is plannable	–5.2pp performance Jun–Aug	Pre-load seasonal capacity
Bottleneck is beds, not triage	79% admit rate at T1 drives breaches	Fix access block / ward flow
Small cohort, big demand	Top 1% of patients = 4.9% of visits	Care-coordination pathway

8. Recommendations

Four actions, each tied to a specific finding and a measurable target. Costs and impacts are illustrative — the point is that every recommendation names the evidence it rests on and the metric that would show it working.

1 · Report NEAT stratified by triage, and fix access block

Replace the blended 4-hour number with a triage-stratified view and set an internal T1–T2 sub-target of **≥80%** (currently ~66.5%). Because admission rate is 79% at T1 versus 3% at T5, direct the operational work at the inpatient bed-access step, not at triage throughput.

2 · Pre-load winter capacity

Period	Presentations	% within 4 hr
Winter (Jun–Aug)	18,229	86.5%
Rest of year	36,608	91.7%

Winter respiratory demand lifts volume and costs **5.2 percentage points** of 4-hour performance. Pre-load a seasonal roster, short-stay/observation beds and a respiratory fast-track from June. A LAG ()-based month-over-month model gives 4–6 weeks of lead time on the ramp.

3 · Roster to the arrival curve

Align senior-clinician and triage-nurse rostering to the hourly arrival curve, targeting the **09:00–13:00** block first — where queueing and LWBS risk compound. No additional headcount required; this is a redistribution of existing shifts.

4 · Care-coordination pathway for frequent presenters

Stand up a care-coordination / Hospital-in-the-Home pathway for the top-1% cohort. A 20% reduction in their avoidable presentations removes **~540 ED visits per year**. Because the same cohort drives the 11.7% readmission rate, one intervention moves two KPIs.

KPI scorecard for ongoing monitoring

KPI	Current	Target	Owner
NEAT % within 4 hr (T1–T2)	~66.5%	≥80%	ED Director
LWBS rate	5.9%	<5%	Nursing Unit Manager
28-day readmission	11.7%	<10%	Discharge / HITH team
Frequent-presenter visits	4.9% of volume	–20%	Care coordination

Appendix — how it was built

Data model

A star schema over three tables: `patients` (demographics, chronic condition count), `ed_presentations` (54,837 rows — arrival mode, triage category, timestamps, disposition, LWBS flag) and `admissions` (11,045 rows — ward, LOS, discharge, readmission link). DDL and constraints are in `sql/01_schema.sql`.

SQL techniques used

Technique	Where it earns its place
Star-schema modelling	Separates the presentation grain from the patient grain, so frequent-presenter analysis does not double-count demographics
CTEs	Each of the 7 business queries reads as a sequence of named steps rather than nested subqueries
<code>LAG()</code>	Month-over-month movement for the seasonal model in §1 and recommendation 2
<code>NTILE()</code> + running totals	Patient value deciles and the Pareto curve in §7
Conditional aggregation	Triage-stratified compliance in one pass instead of five filtered queries
<code>NULLIF</code> -guarded denominators	Rate maths that survives an empty triage-hour cell rather than failing mid-report

Reproducing the analysis

Everything in this document regenerates from the repository in four commands. All randomness is seeded, so every number above reproduces exactly.

Step	Command
1. Generate the synthetic dataset	<code>python data/generate_data.py</code>
2. Build the SQLite database	<code>sqlite3 ed.db < sql/01_schema.sql</code>
3. Run the 7 business queries	<code>sqlite3 -header -column ed.db < sql/02_analysis.sql</code>
4. Render the 14 figures + dashboard	<code>python analysis/eda.py && python analysis/build_dashboard.py</code>
5. Rebuild this PDF	<code>python portfolio/build_ed_report.py</code>

What this analysis cannot tell you

- **The data is synthetic.** It is generated to exhibit realistic ED dynamics and demonstrates method; no conclusion is a claim about any real health service.
- **Access block is inferred, not observed.** The analysis establishes that high-acuity admitted patients drive breaches. Proving the bed-availability mechanism needs ward-level occupancy and bed-request timestamps, which are not in this dataset.
- **LWBS is a flag, not an outcome.** Whether patients who left were subsequently harmed is not answerable from ED data alone; it requires linked follow-up records.
- **The winter effect is described, not decomposed.** Respiratory presentations rise, but separating influenza from RSV from cold-weather injury needs coded diagnosis at a finer grain.

Yizhou Liu · github.com/liuyizhou0402/data-analysis · Full source, SQL, data generator and Power BI build guide in the repository.